INTERIM FEDERAL SPECIFICATION

EXTRACTOR, ELECTRON TUBE (TONG TYPE)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - This specification covers a tong type electron tube extractor used to facilitate the removal of electron tubes from electronic equipment and prevent burning of the operator's fingers. The tool may also be used to install tubes (see 6.2).

1.1.1 Federal specification coverage. - Federal specifications do not include all of the commodities indicated by the titles of the specifications or which are commercially available, but are intended to cover the types, etc., which are generally used by the Federal Government.

1.2 Classification.

1.2.1 Type and size. - This specification covers a tong type electron tube extractor of the size and physical characteristics as hereinafter specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. - The following specifications and standards of the issue in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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Military Specification:
MIL-H-15424 - Hand Tools, Packaging Of.

Military Standard:
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Marking. - Each extractor shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.2 Material. - The two arms of the tong type extractor shall be of steel, SAE 1050 to 1060, hardened and tempered. The material to be vulcanized on the prongs shall be 1/32 inch thick neoprene having a durometer hardness of 40 to 45, or any equivalent oil, grease and heat resistant cushioning material.

3.3 Finish. - The exposed metal surfaces of the extractors, not covered by cushioning material, shall be zinc plated or cadmium plated.

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3.4 Design and construction. - The electron tube extractor shall be of the tong type (see fig. 1 and para. 6.2). The extractor shall consist essentially of two arms with cushioned prong(s) (see 3.2). The two arms shall be riveted together with a 3/32-inch diameter rivet in such manner that the tong type extractor will operate freely, without excessive play, and the finger bows and prongs are in proper alignment. The rivet shall be neatly headed. Each finger bow shall have an inside radius of 9/16-inch, plus or minus 5/64-inch. The arms shall be of 3/16-inch diameter steel, flattened for a distance of at least 7/16-inch on both sides of the rivet hole, to a thickness of 3/32-inch. The outer surface of the flattened portion on each arm shall lie in a plane with the outside of the arm so that when the extractor arms are riveted together the tool will be symetrical. The overall length of the tool, measured in the closed position shall be 8-inches, plus or minus 1/4-inch. The center of the rivet shall be located 4-3/8-inches, plus or minus 1/8-inch from the finger bow end of the extractor. The metal portions of the prongs shall be formed concavely to a 1/2-inch radius for a distance of at least one-inch, measured from the far end of the prong, to a thickness of approximately 3/64 inch. After the cushion grips have been vulcanized in place, and the extractor "opened-up" so the arms are at an included angle of 16 degrees the extractor shall have the prongs in a parallel, or nearly parallel position, and the measurement inside the jaws shall be approximately 15/16-inch (see fig. 1).

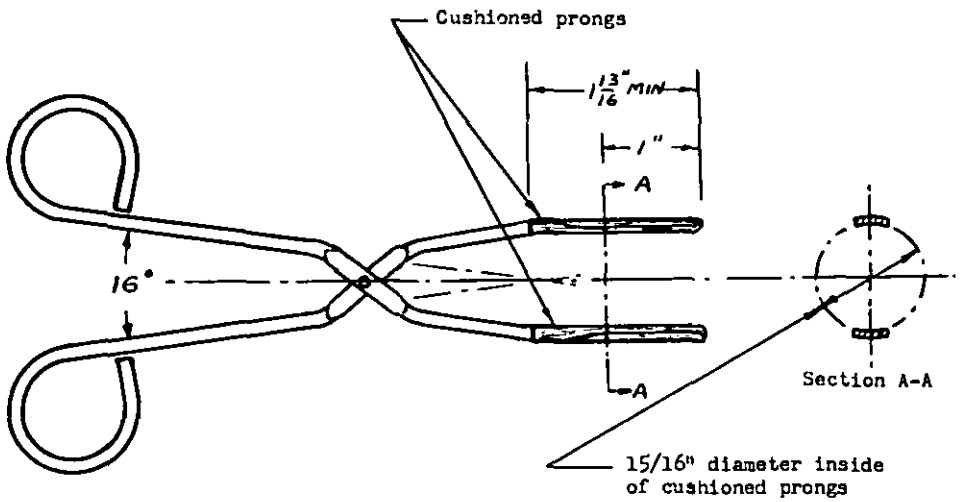


Figure 1. - Extractor, electron tube (see 6.2).

3.5 Illustration. - The illustration shown herein is for the purpose of identification and is not intended to preclude the purchase of electron tube extractors otherwise conforming to this specification.

3.6 Workmanship. - Workmanship shall be first class in every respect. Burrs and sharp edges which might injure the operator shall be removed.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. - All extractors presented at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.3 Sampling for inspection.

4.3.1 Sampling for visual and dimensional examination. - A random sample of extractors shall be selected in accordance with MIL-STD-105 from each inspection lot of material offered for inspection of visual and dimensional characteristics with lot acceptance based on the following sampling inspection requirements in accordance with Standard MIL-STD-105. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 percent defective.

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4.4 Visual and dimensional examination. - Each of the sample extractors selected in accordance with 4.3.1 shall be visually and dimensionally examined to verify compliance with this specification. Examination shall be conducted as specified in table I. Any extractor in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of nonconforming extractors in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected

Table I. - Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Type not as specified.
102	Material(s) not as specified (see 3.2).
103	Finish not as specified (see 3.3).
104	Not in accordance with dimensions specified (see 3.4 and figure 1).
105	The prongs and finger bows not in alignment (see 3.4).
106	The extractor does not operate freely (see 3.4).
107	The extractor has burrs or sharp edges which might injure the operator.
Minor	
201	Marking not as specified.

4.5 Inspection of preparation for delivery. - Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with Military Specification MIL-H-15424, for tweezer, craftsman.

5. PREPARATION FOR DELIVERY

(Civil agencies. - Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified in the contract or order (see 6.2).

5.1.1 Level A. - Level A cleaning, preservation, and packaging shall be in accordance with level A of Specification MIL-H-15424 as specified for tweezer, craftsman.

5.1.2 Level C. - Level C preservation and packaging shall afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements.

5.2 Packing. - Packing shall be level A, B, or C as specified in the contract or order (see 6.1).

5.2.1 Levels A and B. - Levels A and B packing shall be in accordance with levels A and B, respectively, of Specification MIL-H-15424.

5.2.2 Level C. - Level C packing shall insure carrier acceptance and safe delivery at destination. Shipping containers or method of shipment shall comply with rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. - In addition to any special marking required by the contract or order (see 6.1), shipments shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military. - In addition to any special marking required by the contract or order (see 6.1), marking of shipments shall be in accordance with Specification MIL-H-15424.

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6. NOTES

6.1 Ordering data. - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) State the level of packaging and packing required (see sec. 5).
(c) Special marking if required (see 5.3).

6.2 Intended coverage. - This specification is intended to cover electron tube extractors similar to those formerly covered by Signal Corps Drawing SC-D-7459, Tube Puller TL-201.

Notice. - SEE LAST PAGE

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1426 APR 63

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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